山东大学 计算机科学与技术 学院

机器学习 课程实验报告

学号：202200130077	姓名： 薛润泽	邮箱：

实验题目：Experiment 2: Multivariate Linear Regression

实验过程：

（记录实验过程、遇到的问题和实验结果。可以适当配以关键代码辅助说明，但不要大段贴代码。）

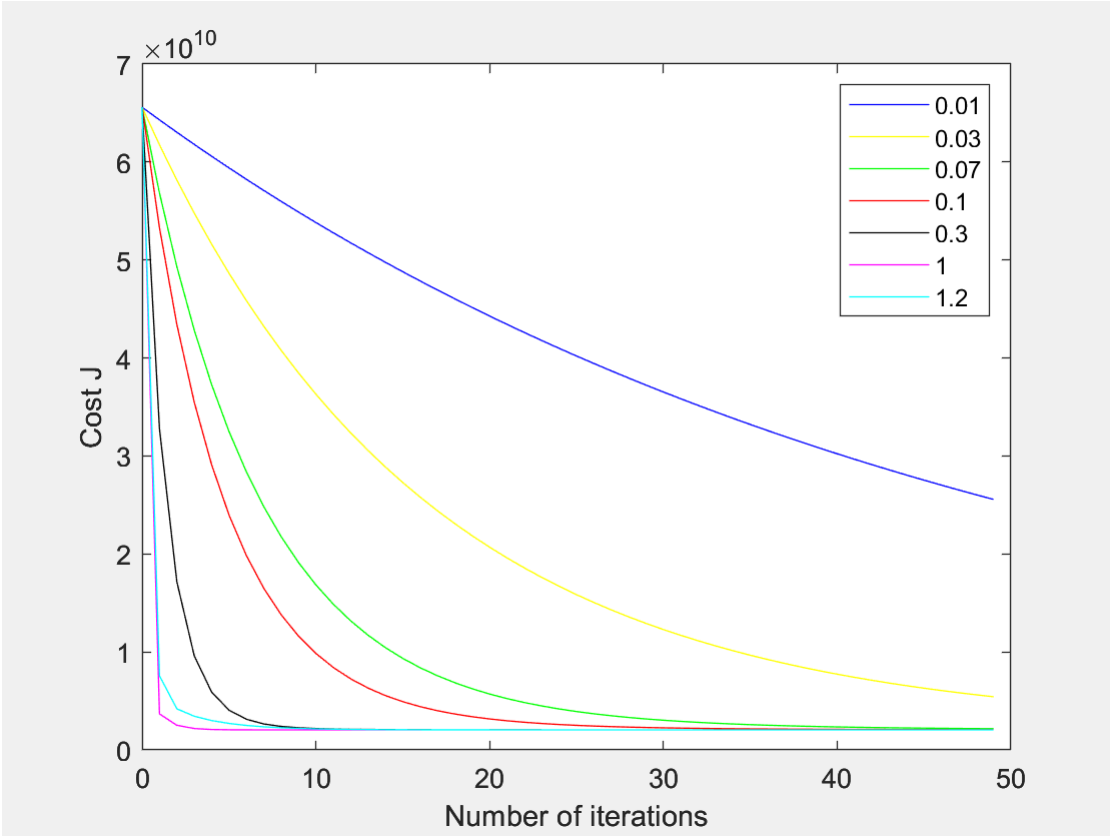
1—3

我们对 x 增加一列数据 1，作为线性回归中的截距项。观察到 x 不同列数据差异过大，我们对每一列计算出其标准差和均值，将其标准化处理。

4

使用梯度下降法来处理线性回归模型，由于梯度下降法需要找到较优的步长 α，所以我们通过记录梯度下降过程中的每一步的损失函数值，观察其值的变化来得到我们需要的较优步长。

通过设置 α 分别为 0.01，0.03，0.07，0.1,0.3，1,1.2，1.5 分别进行梯度下降，并求得每一步的损失函数值，画图展示如下。（下图没有 α = 1.5）



通过观察发现，当 α = 0.1 时，损失函数收敛速度较快且曲线平滑，说明步长为 0.1 时是我们需要的一个较优步长。

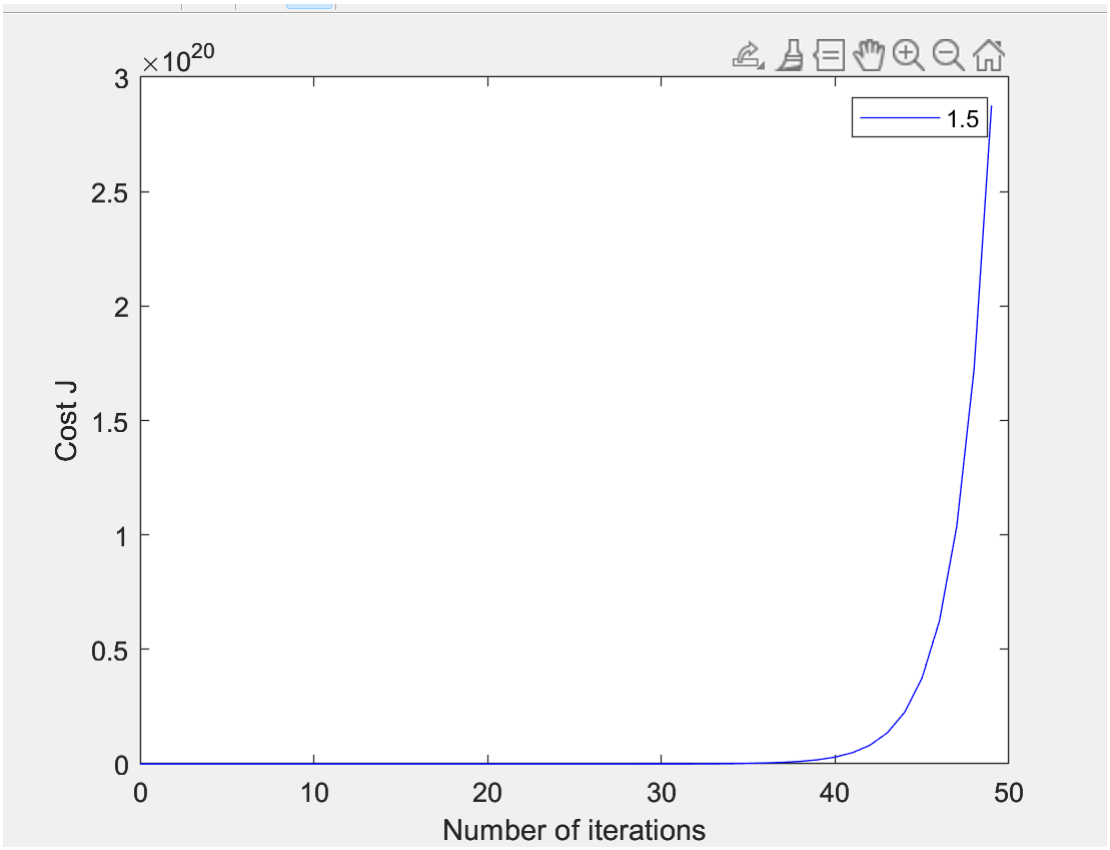
选择 α = 0.1，进行梯度下降法求解 Theta 值，并对测试样本 test = [1,1650,3]进行房价预测。使用训练集得到的均值和标准差对 test 进行标准化

处理。

5.1 当学习率太小或者太大时会发生什么？

观察发现，当学习率太小时每次迭代更新 Theta 的值变化太小，收敛速度过慢，在迭代次数较少时难以达到最优解。甚至在达到最优解前因为 Theta 更新量过小而近乎停止。（如上图 α = 0.01 时，Theta 更新速度小，整体收敛慢）

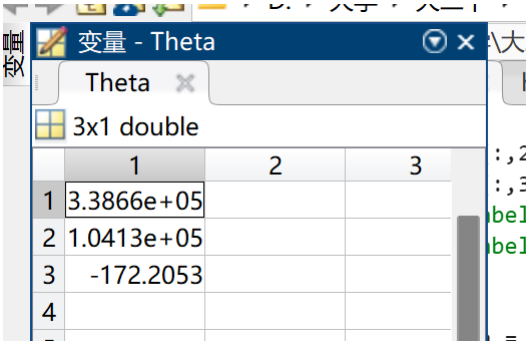
当学习率太大时，Theta 的值更新量过大，可能会在更新过程中越过最优解，导致在最优解两侧反复震荡，导致发散。（如下图 为 α = 1.5 时损失函数的变化，结果发散）



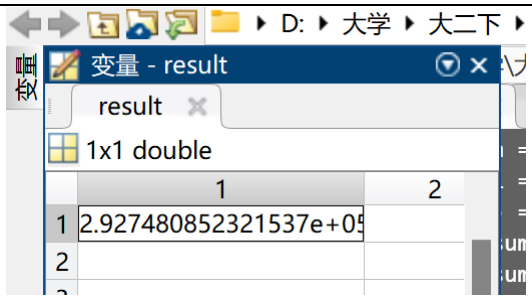
5.2

（a）最优步长为 0.1，在此步长下迭代 50 次后得到 Theta

Theta 为[338658.249249295, 104127.515596898, -172.205333707813]

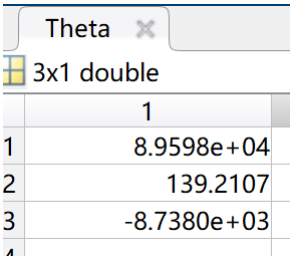


（b）预测结果为 292748.085232154



6.1

Theta 为[89597.9095427972, 139.210674017626, -8738.01911232770]



2 预测结果为：293081.464334896

```
13
14      invtemp = inv(ex2x'*ex2x);
15      Theta = invtemp*ex2x'*ex2y;
16
17      result = test*Theta
18
19
```

```
命令行窗口
>> ex2main

result =

   2.9308e+05
```

观察使用梯度下降法和正规方程解得 Theta 预测的房价，两者非常接近

代码附录：

%函数 hx：

```
function [finalans] = hx(t,ex1x, ex1y,m)
  finalans = 0;
for i = 1 : m
        finalans = finalans + ((ex1x(i,:)*t-ex1y(i))^2);
end
finalans = finalans/(2*m);
```

%画图代码：

```
load("ex2x.dat");
load("ex2y.dat");
[m,~] = size(ex2x);
```

```
ex2x = [ones(m,1),ex2x];
sigma = std(ex2x);
mu = mean(ex2x);

ex2x(:,2) = (ex2x(:,2) - mu(2))./ sigma(2);
ex2x(:,3) = (ex2x(:,3) - mu(3))./sigma(3);

% ylabel('Height in meters');
% xlabel('Age in years');


Theta = zeros(size(ex2x(1,:)))';
J1 = zeros(50,1);
J2 = zeros(50,1);
J3 = zeros(50,1);
J4 = zeros(50,1);
J5 = zeros(50,1);
J6 = zeros(50,1);
J7 = zeros(50,1);
J = zeros(50,1);
%0.01
alpha = 0.01;
for i = 1:50
      J1(i) = hx(Theta,ex2x,ex2y,m);
      sum1 = 0;
      sum2 = 0;
      sum3 = 0;
      for j = 1:m
            sum1 = sum1 +(ex2x(j,:)*Theta-ex2y(j,:))*ex2x(j,1);
            sum2 = sum2 +(ex2x(j,:)*Theta-ex2y(j,:))*ex2x(j,2);
            sum3 = sum3 +(ex2x(j,:)*Theta-ex2y(j,:))*ex2x(j,3);
      end
      sum1 = sum1*alpha/m;
      sum2 = sum2*alpha/m;
      sum3 = sum3*alpha/m;
      Theta(1,1) = Theta(1,1)-sum1;
      Theta(2,1) = Theta(2,1)-sum2;
      Theta(3,1) = Theta(3,1)-sum3;
end
%0.03
Theta = [0;0;0];
alpha = 0.03;
for i = 1:50
      J2(i) = hx(Theta,ex2x,ex2y,m);
```

```
    sum1 = 0;
    sum2 = 0;
    sum3 = 0;
    for j = 1:m
        sum1 = sum1 +(ex2x(j,:)*Theta-ex2y(j,:))*ex2x(j,1);
        sum2 = sum2 +(ex2x(j,:)*Theta-ex2y(j,:))*ex2x(j,2);
        sum3 = sum3 +(ex2x(j,:)*Theta-ex2y(j,:))*ex2x(j,3);
    end
    sum1 = sum1*alpha/m;
    sum2 = sum2*alpha/m;
    sum3 = sum3*alpha/m;
    Theta(1,1) = Theta(1,1)-sum1;
    Theta(2,1) = Theta(2,1)-sum2;
    Theta(3,1) = Theta(3,1)-sum3;
end

%0.07
Theta = [0;0;0];
alpha = 0.07;
for i = 1:50
    J3(i) = hx(Theta,ex2x,ex2y,m);
    sum1 = 0;
    sum2 = 0;
    sum3 = 0;
    for j = 1:m
        sum1 = sum1 +(ex2x(j,:)*Theta-ex2y(j,:))*ex2x(j,1);
        sum2 = sum2 +(ex2x(j,:)*Theta-ex2y(j,:))*ex2x(j,2);
        sum3 = sum3 +(ex2x(j,:)*Theta-ex2y(j,:))*ex2x(j,3);
    end
    sum1 = sum1*alpha/m;
    sum2 = sum2*alpha/m;
    sum3 = sum3*alpha/m;
    Theta(1,1) = Theta(1,1)-sum1;
    Theta(2,1) = Theta(2,1)-sum2;
    Theta(3,1) = Theta(3,1)-sum3;
end
%0.1
Theta = [0;0;0];
alpha = 0.1;
for i = 1:50
    J4(i) = hx(Theta,ex2x,ex2y,m);
    sum1 = 0;
    sum2 = 0;
    sum3 = 0;
```

```
    for j = 1:m
        sum1 = sum1 +(ex2x(j,:)*Theta-ex2y(j,:))*ex2x(j,1);
        sum2 = sum2 +(ex2x(j,:)*Theta-ex2y(j,:))*ex2x(j,2);
        sum3 = sum3 +(ex2x(j,:)*Theta-ex2y(j,:))*ex2x(j,3);
    end
    sum1 = sum1*alpha/m;
    sum2 = sum2*alpha/m;
    sum3 = sum3*alpha/m;
    Theta(1,1) = Theta(1,1)-sum1;
    Theta(2,1) = Theta(2,1)-sum2;
    Theta(3,1) = Theta(3,1)-sum3;
end


%0.3
Theta = [0;0;0];
alpha = 0.3;
for i = 1:50
    J5(i) = hx(Theta,ex2x,ex2y,m);
    sum1 = 0;
    sum2 = 0;
    sum3 = 0;
    for j = 1:m
        sum1 = sum1 +(ex2x(j,:)*Theta-ex2y(j,:))*ex2x(j,1);
        sum2 = sum2 +(ex2x(j,:)*Theta-ex2y(j,:))*ex2x(j,2);
        sum3 = sum3 +(ex2x(j,:)*Theta-ex2y(j,:))*ex2x(j,3);
    end
    sum1 = sum1*alpha/m;
    sum2 = sum2*alpha/m;
    sum3 = sum3*alpha/m;
    Theta(1,1) = Theta(1,1)-sum1;
    Theta(2,1) = Theta(2,1)-sum2;
    Theta(3,1) = Theta(3,1)-sum3;
end
%1
Theta = [0;0;0];
alpha = 1;
for i = 1:50
    J6(i) = hx(Theta,ex2x,ex2y,m);
    sum1 = 0;
    sum2 = 0;
    sum3 = 0;
    for j = 1:m
        sum1 = sum1 +(ex2x(j,:)*Theta-ex2y(j,:))*ex2x(j,1);
```

```
            sum2 = sum2 +(ex2x(j,:)*Theta-ex2y(j,:))*ex2x(j,2);
            sum3 = sum3 +(ex2x(j,:)*Theta-ex2y(j,:))*ex2x(j,3);
        end
        sum1 = sum1*alpha/m;
        sum2 = sum2*alpha/m;
        sum3 = sum3*alpha/m;
        Theta(1,1) = Theta(1,1)-sum1;
        Theta(2,1) = Theta(2,1)-sum2;
        Theta(3,1) = Theta(3,1)-sum3;
end
alpha = 1.2;
Theta = [0;0;0];
for i = 1:50
        J7(i) = hx(Theta,ex2x,ex2y,m);
        sum1 = 0;
        sum2 = 0;
        sum3 = 0;
        for j = 1:m
            sum1 = sum1 +(ex2x(j,:)*Theta-ex2y(j,:))*ex2x(j,1);
            sum2 = sum2 +(ex2x(j,:)*Theta-ex2y(j,:))*ex2x(j,2);
            sum3 = sum3 +(ex2x(j,:)*Theta-ex2y(j,:))*ex2x(j,3);
        end
        sum1 = sum1*alpha/m;
        sum2 = sum2*alpha/m;
        sum3 = sum3*alpha/m;
        Theta(1,1) = Theta(1,1)-sum1;
        Theta(2,1) = Theta(2,1)-sum2;
        Theta(3,1) = Theta(3,1)-sum3;
end

figure;
plot(0:49,J1(1:50),'b-');
hold on;
plot(0:49,J2(1:50),'y-');
plot(0:49,J3(1:50),'g-');
plot(0:49,J4(1:50),'r-');
plot(0:49,J5(1:50),'k-');
plot(0:49,J6(1:50),'m-');
plot(0:49,J7(1:50),'c-');
legend('0.01','0.03','0.07','0.1','0.3','1','1.2')
xlabel('Number of iterations')
ylabel('Cost J')
```

```
%计算使用梯度下降法解得的 Theta  预测的房价
load("ex2x.dat");
load("ex2y.dat");
[m,~] = size(ex2x);
ex2x = [ones(m,1),ex2x];

test = [1, 1650 , 3];


sigma = std(ex2x);
mu = mean(ex2x);

ex2x(:,2) = (ex2x(:,2) - mu(2))./ sigma(2);
ex2x(:,3) = (ex2x(:,3) - mu(3))./sigma(3);

test(:,2) = (test(:,2) - mu(2))./ sigma(2);
test(:,3) = (test(:,3) - mu(3))./sigma(3);
% ylabel('Height in meters');
% xlabel('Age in years');


Theta = zeros(size(ex2x(1,:)))';
J = zeros(50,1);

%0.1
Theta = [0;0;0];
alpha = 0.1;
for i = 1:50
 J(i) = hx(Theta,ex2x,ex2y,m);
    sum1 = 0;
    sum2 = 0;
    sum3 = 0;
    for j = 1:m
        sum1 = sum1 +(ex2x(j,:)*Theta-ex2y(j,:))*ex2x(j,1);
        sum2 = sum2 +(ex2x(j,:)*Theta-ex2y(j,:))*ex2x(j,2);
        sum3 = sum3 +(ex2x(j,:)*Theta-ex2y(j,:))*ex2x(j,3);
    end
    sum1 = sum1*alpha/m;
    sum2 = sum2*alpha/m;
    sum3 = sum3*alpha/m;
    Theta(1,1) = Theta(1,1)-sum1;
    Theta(2,1) = Theta(2,1)-sum2;
    Theta(3,1) = Theta(3,1)-sum3;
end
```

```
result = test*Theta
```

%计算使用正规方程解得的 Theta 预测房价

```
load("ex2x.dat");
load("ex2y.dat");
[m,~] = size(ex2x);
ex2x = [ones(m,1),ex2x];

test = [1, 1650 , 3];

invtemp = inv(ex2x'*ex2x);
Theta = invtemp*ex2x'*ex2y;

result = test*Theta
```

结果分析与体会：

使用梯度下降法和使用正规方程求解线性回归模型最后得到的测试点预测结果相近，而正规方程求解起来不需要进行迭代，也不需要找到一个较优的步长，所以个人感觉正规方程方法更快捷。